

An EXHYTE Agent for Autonomous Scientific Reasoning in Deep-Space Telemetry and Exploration

Executive Summary

Deep-space missions and long-duration low-Earth-orbit (LEO) satellites generate high-dimensional telemetry streams under strict bandwidth, latency, and compute constraints. Recent work has produced strong deep-learning anomaly detectors and on-board autonomy for targeting and health monitoring, but these systems typically optimize single tasks and do not implement a full, closed-loop scientific discovery process. The EXHYTE framework (Exploration–Hypothesis generation–Testing) for AI in scientific discovery explicitly calls out this fragmentation and argues for integrated agents that can explore data and literature, propose testable hypotheses, and design experiments.^{[1][2][3][4][5][6][7]}

This document outlines the research context, gaps, and a proposed architecture for an EXHYTE-based agent that performs autonomous scientific reasoning over deep-space telemetry and related data. The agent targets two intertwined goals: (1) improving spacecraft health understanding and anomaly triage, and (2) extracting scientific insight from unexpected behaviors and observations, all under deep-space operational constraints.^{[8][7][1]}

Background: AI for Space Telemetry and Science Autonomy

AI for spacecraft telemetry anomaly detection

A growing body of work focuses on anomaly detection in spacecraft telemetry using machine learning and deep learning. Approaches include forecasting-plus-threshold models, direct anomaly classification, and image-based encodings of multivariate time series. For example, a 2026 study on the European Space Agency Anomaly Dataset compares forecasting-and-threshold, direct classification, and image-based methods, and optimizes architectures for edge deployment, achieving strong detection performance while fitting within CubeSat-class memory budgets.^{[3][9][5][6][10][11][1]}

Other work uses Gramian Angular Fields (GAFs) to spatially encode multivariate telemetry and applies convolutional networks, showing that stacked multi-channel GAFs capture cross-subsystem behavior and significantly improve event-wise detection metrics on ESA telemetry benchmarks. Additional frameworks leverage LSTM-based models, Gaussian density estimators, and techniques to filter out fake anomalies caused by data errors, some of which are already deployed in ground stations for health monitoring.^{[5][6]}

Autonomous science and targeting in planetary missions

NASA's AEGIS (Autonomous Exploration for Gathering Increased Science) system on the Mars Science Laboratory (MSL) mission is a prime example of on-board science autonomy. AEGIS analyzes images, identifies geologically interesting features, selects targets for the ChemCam instrument, and schedules immediate follow-up measurements, enabling autonomous science without round-trip communication to Earth. This demonstrates the value of integrating perception, prioritization, and action selection on-board, but the system is focused on image-based target selection rather than telemetry-wide scientific reasoning.^{[2][8]}

Broader surveys on AI in space exploration review applications in navigation, mission planning, onboard science, and ground-based analysis, emphasizing the importance of autonomy for missions with high communication delays and limited human supervision. However, most examples deploy specialized modules—navigation controllers, target selectors, or anomaly detectors—rather than a unified discovery loop.^[^12]

AI for scientific discovery and EXHYTE

The EXHYTE framework organizes AI's role in scientific discovery into three phases: Exploration, Hypothesis generation, and Testing. A process-centric survey argues that existing systems often focus on single tasks (e.g., candidate generation or experiment design) and rarely implement closed loops where AI agents iteratively explore data and literature, propose mechanistic hypotheses, and design/execute experiments or simulations to test them.^[^7]

Parallel reports from the U.S. Department of Energy and national laboratories on "AI for Science" emphasize the need for AI that supports full scientific workflows: data management, model building, simulation acceleration, experimental design, and analysis, with uncertainty quantification and rigorous validation. These documents highlight scientific computing for extreme-scale simulations and mission-critical national security contexts but again typically treat modules separately.^{[13][14]}

Problem Statement

Deep-space and deep-LEO missions face three intertwined challenges that motivate an EXHYTE agent:

1. **High-dimensional, heterogeneous telemetry:** Multivariate time series from many subsystems, combined with event logs, commands, and environmental context (space weather, trajectory), produce complex data distributions with rare critical anomalies.^{[1][3][^5]}
2. **Sparse labels and rare events:** True failure or anomaly events are infrequent, and many anomalies are only identified after extensive expert investigation, making supervised training and evaluation difficult.^{[6][3][^5]}
3. **Communication delays and limited bandwidth:** Deep-space missions cannot wait for Earth-based experts to guide every decision; onboard agents must prioritize data, propose explanations, and plan follow-up observations autonomously.^{[4][2][^8]}

Most existing AI systems address anomaly detection, intelligent target selection, or resource-optimized autonomy in isolation. There is a clear gap for agents that perform *scientific reasoning* over telemetry and context, by exploring multi-mission data, generating explicit hypotheses about anomalies or novel phenomena, and testing these via simulation and targeted data acquisition.^{[10][2][3][5][^1]}

The core research question is:

How can an EXHYTE-based AI agent perform closed-loop scientific reasoning over deep-space telemetry—exploring data and knowledge, generating mechanistic hypotheses, and testing them via simulations and follow-up observations—while respecting deep-space operational constraints and safety requirements?

Related Work

Telemetry anomaly detection and health monitoring

Recent work compares deep-learning architectures for spacecraft telemetry anomaly detection and evaluates their effectiveness on benchmark datasets such as ESA’s anomaly datasets and NASA SMAP/MSL data. Techniques include LSTM autoencoders, temporal convolution networks, CNNs on GAF encodings, and hybrid transformer-LSTM ensembles, some of which achieve F1 scores above 0.9 on benchmark datasets and demonstrate feasibility for deployment on space-grade hardware.^{[9][4][5][1]}

Studies also stress practical issues such as irregular sampling, low anomaly density (below 2%), and telecommand-driven state changes that can mimic anomalies, leading to precision/recall trade-offs. Further work examines resource-constrained deployment on CubeSats and FPGA-based platforms, showing that deep networks can be compressed or searched via neural architecture optimization to fit RAM and compute budgets while preserving most detection performance.^{[15][5][6][10][^1]}

However, these systems primarily output anomaly scores or classifications; they do not generate explicit hypotheses about root causes, nor do they plan tests or follow-up observations.^{[3][5][^6]}

Autonomous science operations and target selection

AEGIS on MSL embodies “science autonomy,” where the rover autonomously selects geological targets for further investigation, integrating image analysis, scoring, and command selection into daily operations. This has led to more efficient use of observation windows, especially when communication opportunities are limited. Similar efforts explore AI-driven observation planning, prioritization of downlink data, and resource-aware science scheduling.^{[2][8]}

Nevertheless, AEGIS and related systems focus on local decisions (which rock to target next) and do not perform general hypothesis generation, multi-step testing, or deep integration with telemetry-wide patterns and models of spacecraft behavior. Their design is also instrument-specific, making reuse across missions non-trivial.^{[12][8][^2]}

AI for science and national-scale roadmaps

DOE and NNSA roadmaps for AI in science outline needs for AI systems that can integrate across data sources, simulations, and experimental facilities, accelerating discovery in fields like high-energy physics, materials, and nuclear science. They stress the importance of uncertainty quantification, verification, explainability, and robust software engineering practices for mission-critical scientific applications.^{[16][17][14][13]}

These visions align with EXHYTE's call for closed-loop scientific agents but largely remain at the strategy and prototype level, especially for space missions. Deep-space telemetry introduces additional constraints, including extreme communication delays and the need for onboard autonomy compatible with flight-qualified hardware.^{[14][10][^13]}

Gaps in Current Research

Synthesizing the EXHYTE survey, AI-for-science roadmaps, and space-telemetry literature reveals several key gaps that the proposed agent would target:

1. Task fragmentation vs. closed-loop reasoning

- Telemetry AI focuses heavily on anomaly detection (binary or multi-class) and occasionally root-cause scoring, with limited integration into a full discovery loop.^{[11][5][6][1][^3]}
- Science autonomy systems focus on specific instruments or observation-selection problems but do not unify telemetry-wide reasoning with hypothesis-driven experiment design.^{[8][2][^12]}
- EXHYTE emphasizes the need for agents that connect exploration, hypothesis generation, and testing in one coherent workflow.^[^7]

2. Lack of explicit, mechanistic hypotheses

- Most anomaly detectors output anomaly scores or labels without explicit mechanistic hypotheses (e.g., “reaction wheel bearing friction increase” or “thermal control loop oscillation due to valve latency”).^{[5][6][^3]}
- EXHYTE suggests that hypotheses should be explicit and testable, ideally in symbolic, parametric, or causal forms.^[^7]

3. Limited use of digital twins and simulations for hypothesis testing

- Telemetry models are often trained and evaluated offline; only a subset incorporate digital twins or high-fidelity simulators into the loop.^{[4][10]}
- Some Artemis-related work and hybrid frameworks explore digital twin integration for health monitoring, but they focus on detection performance rather than hypothesis testing and experiment planning.^[^4]
- AI-for-science roadmaps stress the need for coupling AI with simulations and HPC for experimentation, which remains underdeveloped in deep-space telemetry contexts.^{[16][13]}

4. Sparse, noisy labels and rare events

- Benchmark datasets often augment or oversimplify operational realities; rare anomalies, telecommand-driven state changes, and operational transitions make precision–recall evaluation challenging.^{[18][11][^5]}
- There is limited work on agents that actively query experts, propose labeling strategies, or design data-acquisition policies to improve anomaly characterization under labeling constraints.^{[3][7]}

5. Evaluation metrics for scientific value and diagnostic quality

- Most studies report F1, ROC-AUC, or event-wise detection scores; very few quantify improved time-to-diagnosis, reduction in false alarms that consume operator attention, or scientific value of autonomous decisions.^{[6][1][^5]}
- EXHYTE and AI-for-science reports argue for evaluation metrics that capture discovery impact, not just predictive accuracy.^{[13][7]}

6. Human–AI collaboration and explainability in mission operations

- While some work proposes explainable anomaly detection for spacecraft telemetry, systematic frameworks for integrating AI explanations into mission operations and anomaly-review processes remain scarce.^{[19][3]}
- EXHYTE calls for clear models of human–AI division of labor, including when agents should defer to experts and how to present uncertainty.^[^7]

Proposed EXHYTE Agent Concept

High-level objective

The proposed system is an EXHYTE-based agent for autonomous scientific reasoning in deep-space telemetry and exploration. It targets:

- **Operational goals:** Earlier and more informative anomaly detection and diagnosis; better prioritization of downlink and follow-up actions; reduced operator workload.^{[1][5][^6]}
- **Scientific goals:** Extraction of new physical insight from unexpected behaviors and events (e.g., interactions with the space environment, instrument artifacts revealing physics, or rare system responses).^[*]

[*] Scientific-value focus follows the spirit of AEGIS-style science autonomy and AI-for-science roadmaps but is extended to telemetry-wide reasoning.^{[2][8][^13]}

Scope and assumptions

To keep the problem tractable, the initial scope is:

- A single spacecraft or class of spacecraft (e.g., deep-space probe or Earth-orbiting science satellite),
- Focusing on a subset of subsystems (e.g., attitude control, thermal control, power) with available telemetry and basic physics models.^{[5][3]}

- Using a **digital twin or simplified simulator** that captures key dynamics of the chosen subsystems, with parameters that can be manipulated for hypothesis testing.^{[10][4]}

The agent operates in both **offline** (post-facto analysis of mission data) and **simulated online** modes (near-real-time reasoning with streaming telemetry) for research and evaluation.

EXHYTE Decomposition for Deep-Space Telemetry

Exploration layer

The Exploration layer aggregates and organizes information relevant to the mission:

- **Data sources:** Multivariate telemetry time series, event logs, telecommands, mission timelines, environmental context (space weather, trajectory), and historical anomalies.^{[6][1][3][5]}
- **Representation learning:** Unsupervised or self-supervised models (e.g., autoencoders, contrastive learning) learn embeddings of telemetry segments and events, capturing typical modes and rare patterns.^{[9][11][^5]}
- **Event discovery:** Change-point detection and clustering identify interesting segments—sudden changes in subsystem states, correlations across subsystems, or deviations from mission phases.^{[3][5]}
- **Knowledge retrieval:** An LLM-based retriever accesses mission documentation, anomaly reports, and relevant literature, linking patterns in telemetry to known failure modes or physical phenomena.^{[12][7]}

Outputs of the Exploration layer include a ranked set of “events of interest” and associated contextual summaries that feed into hypothesis generation.

Hypothesis generation layer

The Hypothesis layer converts patterns and events into explicit, testable hypotheses:

- **Hypothesis templates:** Domain experts define parametric templates, such as:
 - “Subsystem X exhibits increased friction/drag due to Y,”
 - “Instrument Z’s response curve has drifted by Δ due to radiation damage,”
 - “Observed thermal oscillation is caused by control loop gain parameter deviation.”
- **LLM-based reasoning:** A mission-specific language model takes event embeddings, context, and retrieved documents and instantiates these templates or proposes new ones, with associated confidence and assumptions.^{[13][7]}
- **Causal and structural priors:** Simple causal graphs or system-block diagrams guide hypothesis structure, helping limit the space of plausible causes and avoid physically impossible explanations.^{[16][13]}

Hypotheses are represented in machine-readable form so they can be passed to the testing layer, along with natural-language rationales for human operators.

Testing and experiment-design layer

The Testing layer evaluates hypotheses and plans follow-up actions:

- **Digital twin / simulator coupling:** The agent maps hypotheses to modifications of simulator parameters (e.g., increased friction, degraded sensor sensitivity, altered environmental inputs) and runs ensembles of simulations to generate expected telemetry.
- **Telemetry-simulation comparison:** Statistical distances or likelihood-based metrics compare real telemetry segments to simulated signatures under different hypotheses, updating beliefs (e.g., via Bayesian model comparison or approximate likelihoods).^{[10][4]}
- **Next-observation planning:** Using active learning or Bayesian experimental design, the agent proposes the next best observation or command sequence to discriminate among competing hypotheses, constrained by power, bandwidth, safety, and operational timelines.^{[8][2][13]}
- **Decision support:** The agent produces ranked hypotheses with posterior probabilities, suggested tests, and expected information gain, along with natural-language explanations.

In offline mode, tests are run entirely in simulation. In a future on-board deployment, the experiment-design module would recommend safe, bounded tests or observation strategies for human approval.

Research Questions and Objectives

The proposed project focuses on the following research questions:

1. **Closed-loop performance:** Does an EXHYTE agent improve anomaly explanation quality, time-to-diagnosis, and scientific value compared with standalone anomaly detectors or rule-based workflows?^{[1][5][6][3]}
2. **Hypothesis quality and utility:** Can LLM- and template-based hypothesis generators produce mechanistic hypotheses that are both plausible and discriminable in simulations and follow-up observations?^{[13][7]}
3. **Simulation alignment and uncertainty:** How should the agent represent and propagate uncertainty due to imperfect digital twins, telemetry noise, and unknown external factors?^{[11][4][10]}
4. **Human-AI collaboration:** How should hypotheses, tests, and explanations be presented to operators to support trust and effective decision-making, especially under time pressure and mission constraints?^{[19][3][7]}

The high-level research objectives are:

- Design and implement a prototype EXHYTE agent for deep-space telemetry reasoning on public or simulated datasets.
- Quantitatively evaluate its ability to generate useful hypotheses and tests compared with baselines.
- Propose evaluation metrics that reflect diagnostic and scientific value, beyond anomaly-detection accuracy.

Datasets and Experimental Setup

Telemetry datasets

Candidate datasets include:

- **ESA Anomaly Detection Benchmark (ESA-ADB):** Multivariate telemetry with low anomaly density and realistic operational patterns, used in several recent anomaly-detection studies.^{[5][1]}
- **NASA SMAP and MSL benchmark telemetry datasets:** Used for hybrid deep-learning anomaly detection and digital twin integration; include multiple channels and complex operational regimes.^[4]
- **Open CubeSat laboratory datasets:** Used in resource-aware anomaly detection work, containing thermal and control-system telemetry.^[10]

Where necessary, additional simulated telemetry can be generated using simplified models of spacecraft subsystems.

Digital twin and simulation environment

A simplified digital twin is constructed for the chosen subsystems, with adjustable parameters corresponding to common failure modes and environmental variations. The simulator is integrated as a tool accessible to the EXHYTE agent, enabling automated hypothesis testing.^{[4][10]}

Baselines

Baselines include:

- Standalone anomaly detectors (LSTM autoencoders, CNNs, transformers) trained on the same telemetry datasets, evaluated on detection performance and simple diagnostic outputs.^{[9][11][6][1][5]}
- Rule-based or threshold-based monitoring with manual operator interpretation, to represent current operational practices.^{[6][3]}

The EXHYTE agent's outputs (hypotheses, tests, and prioritizations) are compared against these baselines.

Evaluation Metrics

To align with EXHYTE and AI-for-science goals, evaluation goes beyond pure anomaly-detection metrics:

1. Anomaly detection and triage

- Event-wise F type metrics (e.g., CEF0.5) for anomalies detected and correctly triaged.^{[1][5]}
- Reduction in false alarms compared with baseline detectors and thresholding.^[6]

2. Hypothesis quality

- Precision/recall of hypotheses relative to ground-truth failure modes in simulated scenarios.
- Expert judgment scores (for small studies) on plausibility, usefulness, and clarity of hypotheses.^{[3][7]}

3. Testing and observation planning effectiveness

- Expected information gain or reduction in entropy over hypotheses after proposed tests.
- Number of simulation runs or real observations required to reach a confident diagnosis, compared with scripted procedures.^{[13][4]}

4. Scientific value and operational impact

- Case studies where the agent identifies non-trivial behaviors or correlations that correspond to physical phenomena or operational insights.
- Measures of operator time saved or improvement in mission-planning efficiency in simulated scenarios.^{[2][8][^13]}

5. Robustness and safety

- Sensitivity analyses to simulation mis-specification and telemetry noise.
- Verification that proposed tests respect operational safety constraints and are likely to be acceptable for human approval.^{[16][13]}

Expected Contributions

The project aims to contribute:

1. A concrete EXHYTE instantiation for deep-space telemetry

- A modular agent architecture linking exploration, hypothesis generation, and testing for space-mission data.^{[7][13]}

2. A methodology for telemetry-driven scientific reasoning

- Techniques for mapping anomalies and patterns to explicit, testable mechanistic hypotheses using LLMs and domain templates.^{[3][7]}

3. Integration of digital twins into discovery loops

- A pattern for using subsystem simulators as tools in an AI discovery loop, including uncertainty-aware comparison and experiment design.^{[10][4]}

4. Evaluation framework beyond anomaly accuracy

- Metrics and experimental protocols that measure diagnostic quality, discovery value, and human-AI collaboration in mission operations.^{[19][7][^13]}

5. Open-source prototype

- A research prototype (with synthetic or public data) that can be extended to other missions and domains, aligning with AI-for-science community roadmaps.^{[14][13]}

Future Directions

Once the core EXHYTE agent is validated on a specific subsystem and dataset, future work can explore:

- **Multi-mission transfer:** Training exploration and hypothesis modules on multiple missions or spacecraft classes for transfer learning and generalization.^{[18][5]}
- **Reinforcement learning for experiment design:** Using RL to optimize sequences of tests and observations given dynamic mission constraints.^{[8][2][13]}
- **Tighter integration with ground operations tools:** Connecting the agent with existing mission control and anomaly-review workflows, and designing user interfaces that support human oversight.^{[19][3]}
- **Extension to science instruments:** Applying EXHYTE to instrument data (e.g., spectrometers, imagers) to autonomously generate and test scientific hypotheses about targets and environments, beyond system health.^{[12][2]}
- **Formal verification and assurance:** Developing verification frameworks for EXHYTE-style agents in mission-critical contexts, combining software verification, runtime monitoring, and safety-case arguments.^{[16][13]}

This document can serve as the foundation for a research proposal, thesis, or journal article describing an EXHYTE-based agent for autonomous scientific reasoning in deep-space telemetry and exploration.

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